An Examination of Ankle, Knee, and Hip Torque Production in Individuals With Chronic Ankle Instability

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Abstract

Gribble, PA and Robinson, RH. An examination of ankle, knee, and hip torque production in individuals with chronic ankle instability. *J Strength Cond Res* 23(2): 395–400, 2009—There is some debate in the literature as to whether strength deficits exist at the ankle in individuals with chronic ankle instability (CAI). Additionally, there is evidence to suggest that knee and hip performance is altered in those with CAI. Therefore, the purpose of this study was to determine whether CAI is associated with deficits in ankle, knee, and hip torque. Fifteen subjects with unilateral CAI and fifteen subjects with healthy ankles participated. Subjects reported to the laboratory for one session during which the torque production of ankle plantar flexion/dorsiflexion, knee flexion/extension, and hip flexion/extension were measured with an isokinetic device. Subjects performed 5 maximum-effort repetitions of a concentric/concentric protocol at $60°\cdot s^{-1}$ for both extremities. Average peak torque (APT) values were calculated. The subjects with CAI demonstrated significantly less APT production for knee flexion ($F_{1,28} = 5.40$; $p = 0.03$) and extension ($F_{1,28} = 5.34$; $p = 0.03$). Subjects with CAI exhibited significantly less APT for ankle plantar flexion in the injured limb compared with their noninjured limb ($F_{1,28} = 6.51$; $p = 0.02$). No significant difference in ankle dorsiflexion or hip flexion/extension APT production existed between the 2 groups. Individuals with CAI, in addition to deficits in ankle plantar flexion torque, had deficits in knee flexor and extensor torque, suggesting that distal joint instability may lead to knee joint neuromuscular adaptations. There were no similar deficits at the hip. Future research should determine what implications this has for prevention and rehabilitation of lower-extremity injury. Clinicians may need to consider including rehabilitation efforts to address these deficits when rehabilitating patients with CAI.

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Introduction

Lateral ankle sprains account for a large majority of acute injuries during physical activity (9). Furthermore, the rate of injury reoccurrence and development of lingering instability is also reported to be as high as 80% (28). The contributions of mechanical and functional instability in the lateral ankle complex are often described as chronic ankle instability (CAI) (16). Clinicians are wise to address the neuromuscular deficits associated with CAI in an attempt to return a patient back to function and avoid future incidence of instability. Strength about the ankle is addressed early in the rehabilitation paradigm (17) and is a goal that must be maintained throughout the rehabilitation process. However, clinicians may need to be cognizant of the contributions of strength throughout the lower extremity in patients with CAI and how this may influence the level of functional capabilities (17) and possibly influence the rate of reinjury.

Loss of strength about the ankle complex associated with ankle injury and instability has been studied extensively; however, this relationship does not seem to be consistent. Whereas some authors have reported reductions in inversion and/or eversion strength about the ankle complex (15,23–25,27,29), many authors have reported no significant reductions in force production by the ankle in the frontal (2,7,18,21,22) or sagittal (7,22) planes among those with ankle instability.

Whereas clinicians naturally focus their rehabilitation efforts on the joint complex that has been injured, there may be a need to consider other joints in the lower extremity during the rehabilitation process when faced with ankle pathology. Previous authors have reported alterations in the muscle activation patterns at the hip among those with ankle instability (1,3,4). Additionally, individuals with CAI have been found to have reductions in hip and knee flexion angles and in reach distances during performance of a dynamic postural control task, the Star Excursion Balance Test (12). When assessing the capabilities of a joint, clinicians understand that weakness can be associated with a functional deficit. It seems that functional deficits may be present in the knee and hip of subjects with CAI, but we are unaware of previous studies that have determined whether strength deficits in these joints among this pathological group exist, and knowing this could help explain why functional deficits may exist. This relationship, although not studied extensively, may provide insight on appropriate management of CAI and the associated deficits that may develop proximal to the ankle.

The purpose of this study was to compare the force production capabilities of the ankle, knee, and hip in the sagittal plane among those with and without unilateral CAI. We hypothesized that there would be no deficits in force production at the ankle in subjects with CAI but that there would be deficits in these measures at the knee and hip. If deficits do exist at one or more of these proximal joints, it may be beneficial for clinicians to devote rehabilitation efforts of restoring strength not only at the ankle but also at joints that exhibit reduced force capabilities in their efforts to restore function in patients with CAI.

METHODS

Experimental Approach to the Problem

This study compared sagittal plane torques of the ankle, knee, and hip among subjects with and without CAI to determine whether force production deficits exist at multiple joints in the lower extremity in subjects with this form of ankle pathology. Previous authors have demonstrated altered muscle activation patterns at the hip (1,3,4) and movement patterns during dynamic tasks in the knee and hip (11,12) in subjects with CAI, but no previous investigations have examined whether force production capabilities are diminished in all joints of the lower extremity in these subjects. Therefore, the purpose of this study was to compare the sagittal plane force production capabilities of the ankle, knee, and hip among those with and without unilateral CAI.

Subjects

Fifteen subjects with unilateral CAI (20.3 ± 2.9 years; 1.77 ± 0.1 m; 76.2 ± 13.2 kg; 7 men, 8 women) and without unilateral CAI (23.1 ± 3.9 years; 1.72 ± 0.1 m; 72.7 ± 16.0 kg; 7 men, 8 women) volunteered and signed a university-approved informed consent form. Individuals with self-reported vestibular disorders or head injury in the previous 6 months were excluded from the study. According to a power analysis from a pilot study (10) that examined isokinetic strength of the ankle, knee, and hip in those with CAI, 13 subjects in each group would produce a power level of 0.80. Fifteen subjects were included in each group to exceed this level of calculated statistical power. All subjects were recruited from the university community using flyers and announcements in classes in the department of kinesiology.

Subjects in the CAI group experienced no injury to the lower extremity other than the ankle. They had a history of at least 1 acute lateral ankle sprain, which resulted in swelling, pain, and temporary loss of function (but none in the previous 3 months), with at least 2 episodes of the ankle "giving way" in the past 6 months. Subjects in this group were not participating in a rehabilitation program for their ankle pathology at the time of the study.

All subjects were physically active and participated in at least 30 minutes of continuous exercise 3 times per week. The healthy subjects were free of any lower-extremity injury in the previous year. The healthy group was matched by gender, age, height, body mass, and limb dominance to the members of the CAI group. The dominant limb was designated as the leg that would be chosen when kicking a ball. Additionally, the limbs of healthy subjects were matched to those of CAI subjects by designating a matched "injured" and "uninjured" leg with the same legs of the respective matched CAI subject. For instance, if the injured side of a subject with CAI was the right side, then the matched "injured" side of the matched control subject was also the right side. Matching the subjects for limb dominance and injured side helped to avoid the potential issues of unmatched comparisons of limb dominance that would have occurred with randomizing the limb assignments in the healthy group.

Additionally, all subjects completed the Foot and Ankle Disability Index (FADI) before inclusion in the study. The questionnaire, which assesses self-reported disability and deficits in function of the ankle, consists of 2 parts, the FADI and the FADI Sports Scale, each assessing instability during various forms of daily and physical activities. These instruments have been demonstrated to be sensitive for differentiating between those with and without CAI (14). These instruments have been shown to have high intersession reliability (FADI: ICC (2,1) = 0.98; FADI Sport: ICC (2,1) = 0.94) and moderate to high sensitivity for detecting group and side differences related to CAI (FADI: $p < 0.05$, ES = 0.62; and FADI Sport: $p < 0.05$, ES = 0.95) (14). The results of the FADI and FADI Sport are represented as percentages. Inclusion criteria for the CAI group required a score of < 90% on the FADI and < 80% on the FADI Sport (14). For the subjects recruited in our current study, the CAI subjects scored lower than the control subjects (Table 1) with scores that were consistent with those reported by Hale and Hertel (14) (< 90% on the FADI, < 80% on the FADI Sport), thus demonstrating that subjects were correctly selected and assigned for the 2 groups in this study.

TABLE 1. Foot and Ankle Disability Index (FADI) and FADI Sports Scale (FADI Sport) scores for the chronic ankle instability (CAI) and healthy groups.

	FADI	FADI Sport
CAI	89.3 ± 2.03%	74.8 ± 4.1%
Healthy	100.0 ± 0.00%	100.0 ± 0.00%

Procedures

Subjects reported to the laboratory for a single testing session during which the isokinetic torque production values at the ankle, knee, and hip joints were assessed in the sagittal plane for both limbs. The order of joints and testing limbs was randomized. Group assignment was not blinded, but the assessor was blinded to which limb was the injured side. Subjects were positioned in the isokinetic dynamometer (Kin-Com 500-H, Chattecx Inc.) according to manufacturer guidelines for assessing ankle plantar flexion/dorsiflexion, knee flexion/extension, and hip flexion/extension. The designated movement patterns were assessed with a $60°\cdot s^{-1}$ concentric/concentric protocol. Before each joint assessment, subjects were allowed to practice the movement pattern as many times as they preferred to become familiar with the task. There is some concern about the reliability of sagittal plane isokinetic data acquired from the ankle. However, a recent study using an isokinetic device made by the same manufacturer as used in our study demonstrates strong reliability measures of ankle dorsiflexion (ICC: 0.77–0.93) and plantar flexion (ICC: 0.78–0.95) (22).

After the practice trials, a 2-minute rest period was provided before performing the test trials to avoid potential fatigue. Subjects performed 5 continuous maximal-effort trials of the designated task. When the 5 trials were complete, the subjects were provided a 10-minute rest period before repeating the practice and test trials in the same manner on the contralateral limb. After procedures for the first designated testing joint were completed for both limbs, a 10-minute rest period was provided, and then the procedures were repeated for the second designated joint. Finally, another 10-minute break was provided, and the testing procedures were repeated for the final designated joint.

Statistical Analyses

For each joint direction (ankle plantar flexion, ankle dorsiflexion, knee flexion, knee extension, hip flexion, hip extension), the mean values and standard errors for average peak torque were calculated. Average peak torque was the average of the greatest torque values from each of the individual 5 trials. All torque values were normalized to body mass. For each of these dependent variables, a separate group (CAI, healthy) by side (injured, noninjured) repeated-measures analysis of variance was performed. In the event of a statistical interaction, Scheffe post hoc testing was applied. All statistical analyses were conducted using SPSS 11.0 (Chicago, Ill). Statistical significance was set a priori at $p \leq 0.05$.

Results

Ankle Plantar Flexion

There was a statistically significant group by side interaction for ankle plantar flexion average peak torque (Table 2). A Scheffe post hoc test revealed that for the CAI group, the injured side produced significantly less ankle plantar flexion

Table 2. Average peak torque values for the ankle, knee, and hip in the sagittal plane.

		Average peak torque ($N\cdot m^{-1}\cdot kg^{-1}$) ± *SE*					
Group	Side	Ankle plantar flexion*	Ankle dorsiflexion	Knee flexion*	Knee extension*	Hip flexion	Hip extension
CAI (n = 15)	Injured	0.63 ± 0.06	0.38 ± 0.03	1.11 ± 0.07	1.80 ± 0.12	1.14 ± 0.07	1.75 ± 0.12
	Noninjured	0.77 ± 0.07	0.39 ± 0.02	1.15 ± 0.06	1.91 ± 0.10	1.11 ± 0.07	1.68 ± 0.11
Healthy (n = 15)	Injured	0.70 ± 0.07	0.36 ± 0.03	1.36 ± 0.07	1.89 ± 0.12	1.11 ± 0.07	1.65 ± 0.12
	Noninjured	0.67 ± 0.07	0.33 ± 0.02	1.26 ± 0.07	1.83 ± 0.10	1.08 ± 0.07	1.76 ± 0.11
Group		$F_{1,28} = 0.02$, $p = 0.89$	$F_{1,28} = 1.17$, $p = 0.29$	$F_{1,28} = 2.10$, $p = 0.16$	$F_{1,28} < 0.001$, $p = 0.98$	$F_{1,28} = 0.08$, $p = 0.78$	$F_{1,28} = 0.004$, $p = 0.95$
Side		$F_{1,28} = 2.95$, $p = 0.10$	$F_{1,28} = 0.56$, $p = 0.46$	$F_{1,28} = 0.86$, $p = 0.36$	$F_{1,28} = 0.27$, $p = 0.61$	$F_{1,28} = 0.61$, $p = 0.44$	$F_{1,28} = 0.13$, $p = 0.72$
Group by side		$F_{1,28} = 6.51$, $p = 0.02$*	$F_{1,28} = 1.38$, $p = 0.25$	$F_{1,28} = 5.40$, $p = 0.03$*	$F_{1,28} = 5.34$, $p = 0.03$*	$F_{1,28} = 0.01$, $p = 0.92$	$F_{1,28} = 3.03$, $p = 0.09$

*Highlighted cells indicate a significant relationship ($p < 0.05$).

average peak torque compared with the noninjured side of the CAI group and compared with the matched "injured" side of the healthy group.

Ankle Dorsiflexion

There were no statistically significant relationships for ankle dorsiflexion average peak torque (Table 2).

Knee Flexion

A statistically significant group by side interaction was present for average peak torque (Table 2). A Scheffe post hoc test revealed that the injured side of the CAI group produced significantly less knee flexion average peak torque compared with the matched "injured" side of the healthy group.

Knee Extension

A statistically significant group by side interaction was present for knee extension average peak torque (Table 2). A Scheffe post hoc test revealed that the injured side of the CAI group produced significantly reduced knee extension peak torque compared with the noninjured side of the CAI group and the matched "injured" side of the healthy group.

Hip Flexion

There were no statistically significant relationships for hip flexion average peak torque (Table 2).

Hip Extension

There were no statistically significant relationships for hip extension average peak torque (Table 2).

Discussion

From the results of this study, it seems that CAI is associated with reductions in ankle plantar flexion, knee flexion, and knee extension torque production, whereas deficits in sagittal plane torque production did not exist at the hip or in ankle dorsiflexion. Isokinetic torque production capability is often used as a clinical indicator of strength as well as a patient's ability to progress to functional activities. This result may be a demonstration that CAI is related to deficits in ankle plantar flexion as well as sagittal plane strength at the knee, which may provide clinical information on the sources of limitations to functional capabilities in this pathological group.

Restoration of baseline strength about an affected joint is a rehabilitation goal that is universally followed when treating most musculoskeletal injuries. However, when faced with a chronic instability, the role of strength in the rehabilitation process is not as well defined. Patients with CAI suffer from recurrent bouts of instability at the ankle that are often associated with functional limitations. There is much debate as to whether a significant strength deficit exists in the chronically unstable ankle and whether the potential strength deficit is a significant contributor to the functional and neuromuscular control limitations suffered by these patients.

A large majority of the investigations of strength deficits related to ankle instability have focused on the frontal plane movers of the ankle using either isolated assessments of inversion and eversion or a ratio of torque production between the 2 motions (15,23–25,27,29). However, the results seem to conflict. Some studies report deficits in invertor and evertor strength (15,24), whereas others only report deficits in either evertor (27) or invertor (23,25) strength. Those studies are in contrast to many authors who have reported no significant frontal plane strength deficits in the ankle related to ankle instability (2,19,21,22). In a literature review, Kaminski and Hartsell (18) conclude that there does not seem to be a strong correlation between CAI and frontal plane strength deficits in the ankle.

Although there is much debate revolving around the issue of ankle frontal plane strength deficits in individuals with CAI, there is a paucity of investigations of sagittal plane strength, making conclusions difficult. McKnight and Armstrong (22) report no deficits in ankle sagittal plane strength or work between those with and without functional ankle instability. Our current results contradict this finding, suggesting that deficits in concentric torque production at the ankle exist in plantar flexion, but not dorsiflexion, among sufferers of CAI.

In a previous project using a different group of 30 subjects equally divided into those with and without CAI using the same group inclusion criteria applied in this current study, we observed no statistically significant differences in ankle sagittal plane torque production (10). However, there was a trend observed in that study for plantar flexion ($p = 0.056$) that produced the same significant relationship we have observed in our current results. What is important from the 2 investigations is that clinicians should evaluate and include strengthening for the plantar flexors of the ankle in those with CAI. Additionally, it seems from these 2 investigations that CAI is not associated with deficits in dorsiflexion strength. This relationship may need further investigation because a recent retrospective investigation has suggested that a deficit in dorsiflexion strength may be one of several significant predictors of ankle sprains among physically active men (26).

Although strength deficits are often observed clinically after acute lateral ankle sprains, from these previous studies it seems that there may not be a significant deficit in frontal plane or sagittal plane strength at the ankle that is associated with CAI. Konradsen et al. (20) measured the eversion strength of subjects with first-time acute Grade II and Grade III lateral ankle sprains. A significant evertor strength deficit existed between the injured and noninjured ankles at 3 weeks post-injury but was no longer present at 6 and 12 weeks postinjury. Although it is unclear whether the subjects participated in any formal rehabilitation, it seems that strength deficits related to ankle injury may be relatively short lived, and therefore it is not surprising that strength deficits in the ankle are not consistently observed in patients with CAI in the literature and were only observed in plantar flexion in our results.

There are various measures that may provide insight into the functional capabilities of a joint, one of which is strength or force production. Interestingly, the results of our study suggest that knee strength may be affected by the presence of CAI. This deficit in one aspect of knee function is consistent with previous reports of a deficit in other assessments of knee function among subjects with CAI. Altered patterns of knee flexion during performance of dynamic stability tasks such as jump landing (5,6,13) and the Star Excursion Balance Test (11,12) have been associated with CAI. Additional investigation is needed to determine whether there is a relationship between this kinematic pattern and the reduction in knee strength as suggested by the results of our current study.

Our current results demonstrate deficits in knee flexion and extension torque production in subjects with CAI. This is consistent with the pilot study previously described in which subjects with CAI also demonstrated deficits in knee extension strength (10). Deficits in knee strength are often associated with other pathologies such as anterior cruciate ligament rupture and patellofemoral pain syndrome. However, this is one of the first investigations to document that similar strength deficits are associated with a pathology not directly affecting the knee. Future investigations will need to quantify the muscle activation patterns of the muscles controlling the knee to determine whether these relationships do exist and are creating any limitations during functional activities in those with CAI. Additionally, it is unknown whether this isolated reduction in force production at the knee is a causative or a lingering factor that contributes to CAI. It will be important to ascertain the source and impact of these relationships if clinicians are to use this information to prevent and rehabilitate neuromuscular deficits after CAI more efficiently.

The results of our study did not yield a significant difference in hip flexion and extension torque production between those with and without CAI. This was somewhat unexpected to us because previous investigations have produced contradictory results (10). Additionally, previous authors have reported deficits in hip activation patterns among those with CAI (1,3,4,12).

Previous authors have attempted to explain why changes in neuromuscular control are observed in the presence of a distal joint instability at the ankle during the performance of functional activities. Beckman and Buchanan (1) hypothesize that the pattern of improved hip muscle recruitment during a dynamic activity may be the result of deafferentation of peripheral nerves in the ankle, which created a shift from an ankle strategy to a hip strategy in an effort to maintain postural control. However, they explain that these responses in hip muscle recruitment are too slow to be a central nervous response to an afferent signal and may indicate that, with peripheral joint injury, a centrally mediated feed-forward mechanism could recruit proximal muscles to protect against injury at the distal joints. Similarly, Caulfield and Garrett (5) have suggested that the observed kinematic changes in the knee of a limb that has suffered ankle instability during a landing task may indicate the creation of a protective feed-forward mechanism.

However, in our current investigation, the measures of isolated joint performance through torque production occurred at a much slower rate than typically occurs during functional movements and were only designed to require the use of a single muscle group. It may be difficult to conclude that a feed-forward mechanism would explain why the force production deficits were observed at the knee, because the isokinetic muscle contractions should afford a subject adequate time to plan the movement without requiring the need for a faster feed-forward mechanism to enable the necessary efferent response. Additionally, it has been suggested that a reduction in torque production may be indicative of a reduction in neural drive (8). It is possible that the existing instability at the distal joint has influenced neural drive to the knee, lending further support to the possibility that CAI may lead to reorganization within the central nervous system that results in altered proximal joint neuromuscular control. However, without electromyography or a measure of voluntary activation using a superimposed electrical stimulation (which were unavailable in this experiment), it is unknown whether the muscles were recruited differently between the groups in our study and whether that could help explain the differences in force production.

Although the inclusion criteria for our CAI group were defined specifically, and very similar criteria have been used in previous investigations (11,12), universally accepted inclusion criteria for this pathology do not exist. Because of the inconsistencies across the literature, there is a need to develop a consensus in the clinical and research communities on the description of this pathology so that a consistent set of inclusion criteria may be established. This will allow for more appropriate comparison and application of results on the influence of CAI.

In this study, we selected isolated assessments of concentric average peak torque values for the sagittal plane movements of the ankle, knee, and hip. Future studies should investigate these same relationships at more velocities than those examined in this study and also during eccentric contraction assessments. Additionally, several studies have suggested the importance of using inversion/eversion torque ratios in determining the influence of CAI on ankle strength (15,22,24,29). It will be useful in future investigations to incorporate similar ratios between the sagittal plane movement patterns that we included in this study to gather additional information on the role of CAI on lower-extremity strength. Finally, to completely elucidate the importance of strength deficits in the lower extremity on the management and rehabilitation of CAI, future investigations should include rehabilitation interventions designed to isolate and overcome the strength deficits along with measures of functional activity to determine whether these therapeutic efforts are efficient at improving the quality of life and level of activity in patients with CAI.

To our knowledge, this is the first investigation to examine the sagittal plane force production capabilities throughout the lower extremity in sufferers of CAI. Our results demonstrate that even though the source of instability existed at the ankle, the deficits in sagittal plane force production were observed at the ankle and the knee. This finding is important for clinicians as they consider new methods for approaching rehabilitation for ankle instability, and it suggests that efforts may need to be focused at the knee in addition to the ankle when progressing these patients throughout the therapeutic rehabilitation process.

Practical Applications

Isokinetic torque production capability is often used as a clinical indicator of strength as well as a patient's ability to progress to functional activities. This result may be a demonstration that CAI is related to deficits in sagittal plane strength at the knee and ankle, which may provide clinical information on the sources of limitations to functional capabilities in this pathological group. However, clinicians should approach these results with some caution because the group differences in ankle and knee joint torque, although statistically significant, could be considered small, with an average difference of approximately 0.10 N·m. Strengthening is an important part of the rehabilitation process, and this information may lead to new approaches to rehabilitation for patient with CAIs and might lead clinicians to develop more efficient approaches to addressing this pathology. Whereas these results may provide information about the potential differences in strength at the ankle and knee in patients with CAI, future investigations should continue to examine deficits in strength and other factors that will dictate successful progression to the return of functional status in those with CAI.

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